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Safety

BluePrint Automation is not and does not represent itself as a safety expert. It is the responsibility of the owner, employer, or equipment user to take all necessary steps to guarantee the safety of all personnel in the workplace. According to **ISO 12100** *Safety of Machinery***,** the owner or user is advised to consult the latest standards to ensure complete compliance to safety Requirements for packaging machinery, packaging-related converting machinery, and automated systems. As the owner, employer, or user of a packaging machine, it is your responsibility to arrange for training of operator(s) and maintenance of machinery to recognize and respond to known hazards associated with your machinery and to be aware of the recommended operating procedures for your particular application and machinery installation. **BluePrint** Automation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the machinery be trained in an approved **BluePrint** Automation training course and become familiar with the proper operation of the system. Persons responsible for programming the system must be familiar with the design, implementation, and debugging of application programs.

Safety

Signal Word Panels

The convention followed in this manual for signal word panels comports with international standard **ISO 3864** *Graphical symbols -- Safety colours and safety signs*. The signal word panels are deployed as follows:

NOTICE — This signal word panel is used to communicate information about practices not related to personal injury.

⚠ CAUTION — This signal word panel indicates a hazardous situation which, if not avoided, ***could*** result in **minor** or **moderate injury**.

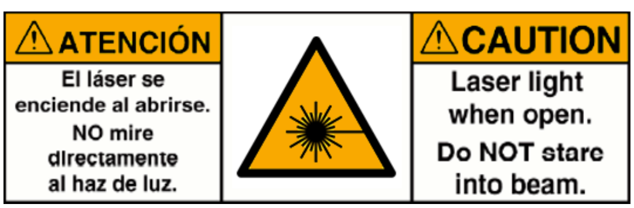
⚠ WARNING — This signal word panel indicates a hazardous situation which, if not avoided, ***could*** result in **death** or **serious injury**.

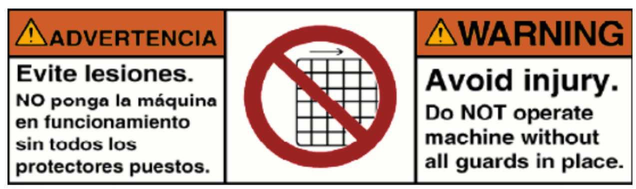
⚠ DANGER — This signal word panel indicates a hazardous situation which, if not avoided, ***will*** result in **death** or **serious injury**.

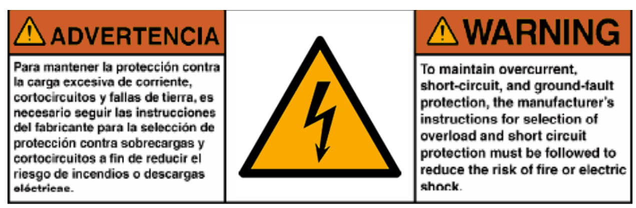
Safety Signs

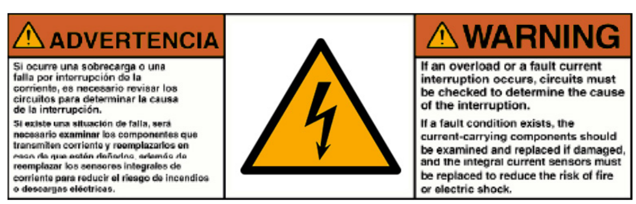
Safety signs are placed on this machine as a constant reminder that the operator must be sharp and alert when operating this machine. Never remove, paint, or deface a safety sign. Should a safety sign be damaged or defaced, contact **BluePrint** Automation for a replacement.

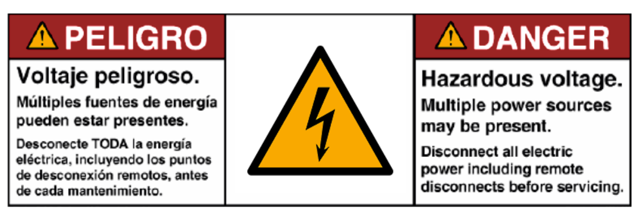
The following safety signs are used on **BluePrint** Automation machines. Review the signs that are on your specific equipment and be familiar with their locations and messages.

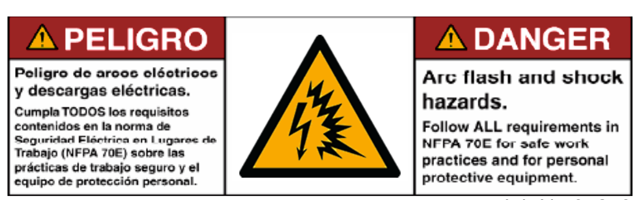












General Safety Precautions

Please read and follow all of the safety precautions listed below before operating the machine loading system.

BluePrint Automation machines are designed and manufactured following high quality and safety standards. We have given extra attention to easy operation and effective hazard protection for personnel. However, all equipment can become a hazard when operated incorrectly or without the necessary training and maintenance.

This chapter is not a complete list of all possible dangers. Specific working conditions cause specific hazards. The purpose is to make you aware of possible risks and to encourage safe working conditions at and around the machine. If you do not obey safety instructions, this can cause personal injury and / or damage to products and equipment.

- Failure to use, maintain, or service the equipment in accordance with this Operation and Technical Reference Manual and the associated vendor-supplied documentation may impair the safety protection provided by the system.
- A shock hazard exists within the equipment control enclosure(s). There are some exposed 480 V~ and 230 V~ connexions within the enclosures. A qualified electrician must secure access to the electrical control enclosure(s) before attempting to operate on, troubleshoot, or repair problems with the system.
- Use one of the emergency stop push-buttons in case of emergency when injury to personnel or damage to equipment is imminent.
- Observe all safety precautions at all times to reduce the chance of serious injury or equipment damage
- Wear safety glasses at all times around operating equipment. Check all pneumatic (if applicable), electrical, and mechanical connexions periodically for tightness and signs of wear. Replace or tighten, as required, to maintain the safety and integrity of the system.
- Do not operate the packaging machinery system with any of the safety enclosure panels removed. Panels that require a tool to remove are not protected by safety switches.
- Keep your hands clear of the rotating parts and conveyor rollers. There is a risk of getting your hands or fingers pinched in the mechanisms.
- The system includes safety mechanisms to protect the operator from inadvertently placing hands or equipment into the moving parts during operation. If the associated sensors or switches are triggered, an emergency stop will result.

It is not permitted to modify the machine without prior written permission of **BluePrint Automation**. As soon as a part is replaced with an out of specifications part, this is a modification, especially electronics and control parts. As long as there is no change in power, the replacement can be classified as a repair and this is permitted. If a modification is made without permission of **BluePrint Automation**, our legal liability automatically expires.

⚠ WARNING

Use extreme care around the electrical components, and any moving parts of the machine because of the risk of personal injury.

⚠ WARNING

Do not disconnect, defeat, or remove any of the safety devices installed on the machine. Under normal operating conditions, these devices protect the operator and technician from moving or electrical parts. The safety devices include the emergency stop push-buttons, lockable disconnect switch, safety switches, safety relays, fixed guards, and openable guards.

DANGER

While the machine is powered up.
- ***Do not lean against or on the machine.***
- ***Do not reach into the machine.***
- ***Do not climb onto the machine.***
- ***Do not crawl under the machine.***
- ***Do not put objects on the machine.***

WARNING

- ***Always keep sufficient distance from all moving parts.***
- ***Avoid loose clothing, ties or loose long hair. Always put on working clothes, working boots and safety glasses. When necessary, take additional safety precautions.***
- ***Only use the machine for the purpose for which it has been designed.***
- ***Obey the safety policies of your company.***
- ***Read the safety instructions of integrated equipment, refer to the OEM documentation.***

NOTICE

Not all equipment manufacturers use the same warning signs are on their equipment.
For integrated equipment, also refer to the OEM documentation.

Mechanical Safety

Allow enough area around the machine for operating and cleaning personnel access. Once this equipment is set, permanently fasten the support pads securing the equipment to the floor. Visually check the interior and exterior of this equipment to ensure that all shipping and transport restraints have been removed. Electrical wiring and pneumatic tubing are enclosed inside the frame tubing. Never drill into or weld the frame.

Physical Barriers

Because of its rotating and fast-moving parts, our machines are protected on all sides, usually by fixed or moveable transparent doors and panels. Positionable guards are fitted with safety switches that stop the machine when the guard is opened. The product/material in-feed and out-feed openings understandably cannot be enclosed and herein lies the danger. Do not insert any limbs in the product/material in-feed and out-feed openings while power is applied to the machine.

Operator Guards

When operating this machine all guards must be closed and in their proper locations.

Before operating this machine do the following:

- Check for loose guards.
- Check for missing guards
- Check for broken guards.
- Tighten or replace guards as necessary.

Do not alter or modify any guard.

Physical Contact

When the machine is powered (i.e., not in **Lockout** condition) remember the following:

- Do not lean on the machine
- Do not reach into the machine
- Do not climb on the machine
- Do not crawl under the machine

Minor Maintenance

Place the main power disconnect switch in the Off position and **Lockout** condition before cleaning, adjusting, or performing maintenance. Before returning the main power disconnect switch in the On position, make certain all guards have been returned to their proper location and are closed.

WARNING

- ***If the machine has feet, never extend the feet of the machine too far. This can cause the machine to become unstable or fall.***
- ***Make sure the machine guards are sufficiently close to the floor, even if the floor is uneven. These guards must always prevent persons from reaching into the machine.***
- ***Do not start the machine if it has a defect.***
- ***Never allow the electrical interlocks to be bridged or put out of service.***
- ***Do not modify the electrical or mechanical guards without the written consent of BluePrint Automation.***
- ***Some pneumatic parts remain pressurized. Wait until all moving parts in the section you want to reach have stopped.***
- ***Do not insert limbs or objects into the in-feed and out-feed openings when the machine is in operation.***

Electrical Safety

Safety Switches

Safety switches are used to render the machine inoperable should a safety panel or guard be opened or removed. Safety switches provide increased protection for operators. For your protection do not override, defeat, or bypass safety switches.

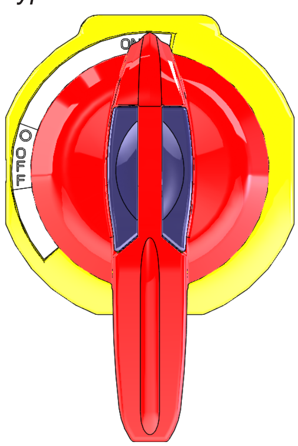
Electrical Panels

Keep electrical power panel enclosures closed at all times. Allow only authorised personnel access into electrical panel enclosures.

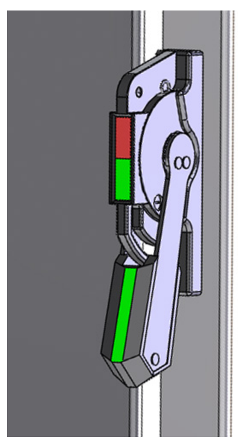
Electrical Supply

Interconnecting plugs and cables are used to install this equipment. Keep all cables off the floor. The main electrical cabinet has a disconnect handle. Placing this handle in the Off position deënergises the equipment. However, the input side of the disconnect switch is still energised. Turn off and **Lockout** the main power source to the cabinet before servicing.

Typical Main Power Switches



Rotary Type



Lever Type

NOTICE

Electrical Supply: Because our clients have different requirements, gland access holes for installation of the main electrical power are not provided. Be certain wire gauge and insulation meet all applicable code requirements. The power source must have a breaker in a lockable enclosure.

⚠WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

Emergency Stop Push-button

All machines are equipped with emergency stop push-button(s). If an emergency occurs, the machine or zone can be shut down by simply pressing any one of the emergency stop push-buttons. When an emergency stop push-button is pressed, all motors and belt conveyors lose driving power. In addition air pressure on all pneumatic parts is removed and a service stop signal is transmitted to upstream equipment.

Emergency Stop Push-button

Do a test of all emergency stops at least once a month to make sure that safety-related features operate correctly.

Mechanical Installation Inspection

Allow enough area around the machine for operating and cleaning personnel. Once this equipment is set, permanently fasten the support pads securing the equipment to the floor. Visually check the interior and exterior of this equipment to ensure that all shipping restraints have been removed. Electrical wiring and pneumatic tubing are enclosed inside the frame tubing. Never drill into or weld the frame.

When inspecting the machine, be sure to:

- Turn off power at the controller.
- Lockout and Tag-out the power source at the Main Power Switch according to the policies of your plant.
- If machine motion is not needed for inspecting the electrical circuits, press the Emergency Stop Push-button on the operator panel.
- Never wear watches, rings, neck-ties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the machine motion or electrical circuits, be prepared to press the Emergency Stop Push-button, in an emergency.

⚠ WARNING

Never bypass, strap, or otherwise defeat a safety device such as a limit switch, for any operational convenience. Defeating safety devices is known to have resulted in serious injury or death.

Maintenance Safety

⚠ WARNING

Do not try to remove any mechanical component from the machine before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

Know the work area of the machine and *all* guard hoods and/or door opening devices.

- Know the location and status of all switches, sensors, and/or control signals that might cause the machine elements, conveyor, and opening devices to move.
- Make sure that the work area near the machine is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the machine will perform *before* operating the machinery.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neck-ties, scarves, or loose clothing that could get caught in moving machinery.
- Stay out of areas where you might get trapped between moving conveyors, or an opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of audible alarms.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.
- When a maintenance technician is repairing or adjusting a machine, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the **Emergency Stop Push-button**. This person must be knowledgeable of the machine and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Never use machine power to aid in removing any component from the machine.
- During machine operations, be aware of the moving elements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before undertaking cleaning operations.

Conveyor Safety

- When equipment is interfaced between **BluePrint Automation** equipment and existing equipment, care needs be to paid to the guarding of the abutting equipment.
- If conveyors are installed above exits, passageways, aisles or any other place where people walk, a clearance of at least 6 ft, 8 in shall be provided. If it is necessary to install them with less than the minimum requirement then an alternate exit shall be provided. Also, a warning of low headroom must be posted.
- Only trained employees shall be allowed to operate conveyors. Training must include operation for normal and emergency situations.
- Keep the area around the conveyors clear of debris and obstructions.
- When working near any conveyor:
- Wear hard hat and safety shoes.
- Tie back (and tuck in) long hair.
- Know the location of the emergency "shut-off" devices and how to use them.
- Do not wear loose clothing or jewellery.
- Do not climb on the conveyors.
- Position yourself so that you are not hit by objects moving down the conveyor.
- Ensure that you can see the conveyor system when you are at the operating controls.
- Ensure that guards are in place for all moving parts of the drive system and in all zones where hazards such as in-running nip, drawing-in, trapping and crushing, friction burns or abrasion are present.
- Locate emergency stop devices near the operator and along the length of the conveyor at approximately 30 metres (100 feet) apart (or closer).
- Ground belts on belt conveyors to prevent static build-up.
- Gravity conveyors include those that have rollers, wheels or chutes where objects move by gravity or momentum only.
- Guard pinch points on rollers and wheels and between the conveyor and receiving table.
- Provide adequate guardrails along the sides to prevent all objects from falling off.
- Provide retarders (friction areas) if heavy objects are conveyed.
- Ensure there are warning devices near the receiving areas if you cannot see the packages moving on the conveyor.
- Ensure draft checks (fire doors) are installed where conveyors pass through fire walls or floors.

⚠ WARNING

Never ride on, step on, or rest on a conveyor system.

⚠ WARNING

Do not climb, sit, stand, walk, ride, or touch the conveyor at any time.

⚠ WARNING

Reference the OEM supplied conveyor manuals for additional safety information related to the various conveyor systems as applicable.

It is imperative that workers never climb, sit, stand, walk, ride, or even touch the conveyor line at any time. It's common sense, but people tend to get mischievous about it and there are injuries and equipment damage every year due to not following this rule.

- Operate equipment only with all approved covers and guards in place.
- Do not load a stopped conveyor or overload a running conveyor.
- Ensure that all personnel are clear of equipment before starting.
- Allow only authorised personnel to operate or maintain material handling equipment.

⚠ WARNING

Do not perform maintenance on the conveyor until electrical, pneumatic, hydraulic, and gravity energy sources have been blocked or placed into Lockout condition.

⚠ WARNING

Keep clothing, body parts, and hair away from the conveyor.

Only workers who have been trained should be permitted to operate and perform maintenance on conveyors. This is for two reasons:

1. Safety of the technician: Conveyors can be dangerous to those who do not thoroughly understand the equipment and how to safely work on it.

2. Only trained personnel can reliably maintain a conveyor to perform at peak efficiency. This isn't necessarily a safety concern, although it can be if sub-par conveyor performance causes workers to try to look at it on their own or bypass guards.

- Do not modify or misuse conveyor controls.
- Remove trash, paperwork, and other debris only when power is in Lockout/Tag-out condition. Ensure that ALL controls and pull cords are visible and accessible.
- Know the location and function of all stop and start controls.
- Report all unsafe conditions. Jams should be cleared ONLY BY authorised, trained personnel.

Installing/Removing Lockout/Tag-out Devices

Installing Lockout Devices

The following general guidelines, while not a substitute for a well-developed internal **Lockout/Tag-out** policy, should be observed if maintenance or servicing is required around high-voltage components or circuits or on pneumatically actuated equipment:

1. Place electrical power under **Lockout** condition using approved **Lockout** devices. Refer to your company's internal policies for specific **Lockout/Tag-out** procedures to be followed for placing equipment under **Lockout** condition during maintenance and service activities.
2. Tag the **Lockout** devices using authorised **Lockout/Tag-out** procedures and tags.

Removing Lockout Devices

1. Comply with internal procedures for the removal of **Lockout** devices.
2. Resolve the situation that required the energy sources to be placed under Lockout condition.
3. Make sure the machine and work area are clear of all tools and equipment.
4. Remove **Lockout** device(s) from the pneumatic shut-off valve(s).
5. Rotate the pneumatic shut-off valve handle to the **ON** position.
6. Remove the lockout device from the Electrical Disconnect switch(es).
7. Rotate the Electrical Disconnect switch to the **ON** position.

DANGER

If it should ever become necessary for personnel to access a powered electrical cabinet, it is the end-user's responsibility to implement an Arc-Flash safety policy in compliance with all relevant safety regulations and legislation in effect in the end-user's geographic location.

Energy Lockout

BluePrint Automation machines are suitably predisposed for placing all energy sources in **Lockout** condition using approved energy blocking devices. Refer to your company's internal procedure for the specific program to be followed for locking out energy sources during maintenance and service activities.

Machine Elements with Lockout Provision

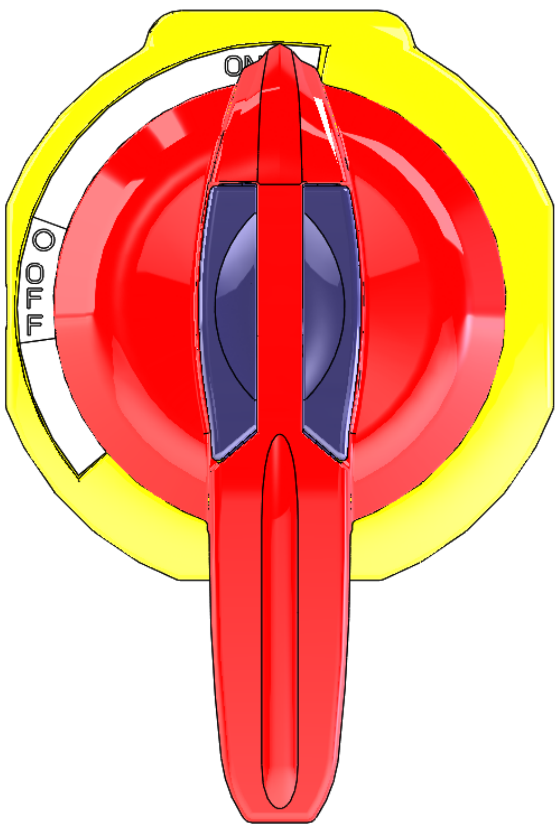
Main Power

BluePrint Automation uses two different types of Main Power Shut Off Switches. One switch is rotary-actuated (A) and the other is lever-actuated (B). ***Refer to the section that matches your machine.***

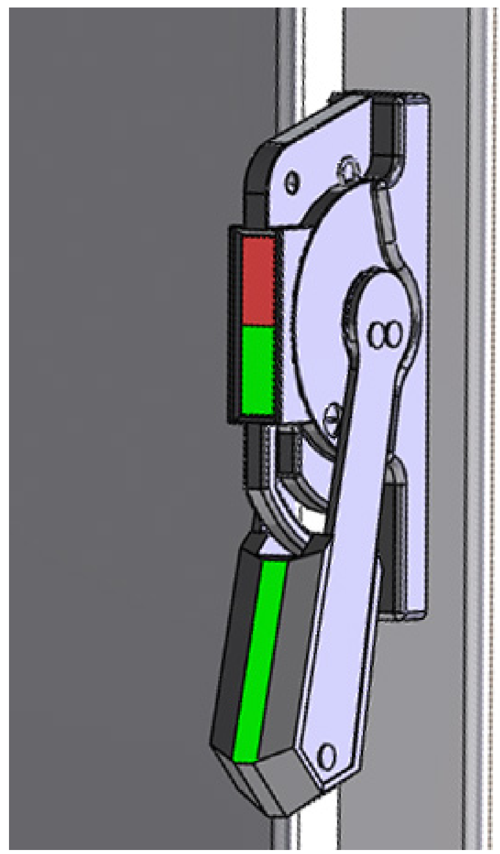
NOTICE

It is the end-user's responsibility to implement a Lockout/Tag-out policy in compliance with all relevant safety regulations and legislation in effect in the end-user's geographic location. Lockout/Tag-out refers to an energy-control program to ensure that employees isolate machines from their energy sources and render them inoperative before any employee services or maintains them.

A - Rotary-Actuated

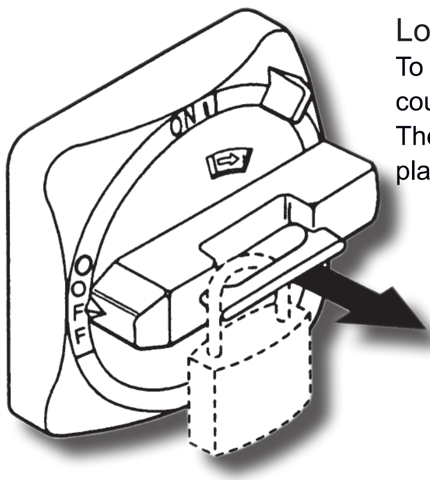


B - Lever-Actuated



1. To place all electrical power into Lockout condition, move the Power Disconnect to the **OFF** position and align the holes for the lockout device installation. Install an authorised lockout device in the holes so the switch is mechanically blocked from being moved to the **ON** position.
2. Tag the electrical disconnect switch using authorised internal procedures and tags.

When the main power switch is turned off, all components and accessories associated with the system will lose power, (unless special wiring was specified by your company at the time of purchase, in this case, the cables will have yellow jackets).

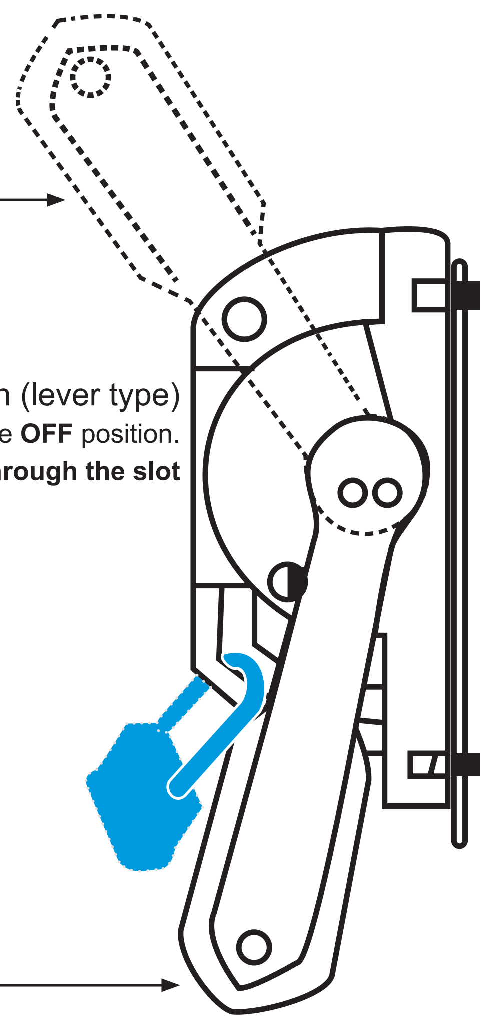


Lockable Main Power Switch (rotary type)

To place the Main Power into **Lockout** condition, turn the red switch handle counter-clockwise to the OFF position.
The Main Power is now OFF. Pull out the black centre tab of the handle, and place a lock through the centre opening.

Locked Out
Main Power Switch

ON Position



Lockable Main Power Switch (lever type)

Move the handle from the on position to the **OFF** position.
The Main Power is now OFF. Place a lock through the slot

OFF Position

Laser Safety

This system incorporates a laser product, the safety information for which is detailed below.

Class 1 Lasers

An IEC 60825-1 Class 1 laser is safe under all conditions of normal use. This means the maximum permissible exposure (MPE) cannot be exceeded when viewing a laser with the naked eye or with the aid of typical magnifying optics (e.g., telescope or microscope). To verify compliance, the standard specifies the aperture and distance corresponding to the naked eye, a typical telescope viewing a collimated beam, and a typical microscope viewing a divergent beam. It is important to realize that certain lasers classified as Class 1 may still pose a hazard when viewed with a telescope or microscope of sufficiently large aperture. For example, a high-power laser with a very large collimated beam or very highly divergent beam may be classified as Class 1 if the power that passes through the apertures defined in the standard is less than the accessible exposure limits (AEL) for Class 1; however, an unsafe power level may be collected by a magnifying optic with larger aperture.

Laser Specifications

Divergence angle: 1 mrad; wavelength: 655 nm; frequency: 7.1 kHz; pulse duration: 0.2 μs; limit value pulse: ≤ 140 mW (IEC 60825-1).